

 OR-AS Operations Research Applications and Solutions	Case Name: Urban Road Project Owner: Eder Omar Vilca Carrillo	Sector	Construction
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule	Schedule with resources
			Schedule with costs
		Risk Analysis	Random simulation
			One of nine std. scenarios
Processed by	Amin Azimi		User defined distributions
Date	2025	Project Control	Automatic tracking
File Name	C2025-02		Tracking based on user input

1. Project description

Project authenticity

The construction of road pavement, sidewalks (veredas), and culverts to improve urban mobility and stormwater management.

The project consists of activity and cost data that were obtained directly from the actual project owner

2. Project properties

2.1. Baseline Schedule

General		Network topology	
# Activities	100	Serial/Parallel (SP)	30%
Planned Duration (PD)	52 days	Activity Distribution (AD)	41%
Budget At Completion (BAC)	197,472.31€	Length of Arcs (LA)	0%
Renewable Resources	6 Types	Topological Float (TF)	40%
Consumable Resources	0 Types		

* standard eight-hour working days

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	10.13	8.98	0.83
CRI-rho	16.83	15.25	1.17
CRI-tau	20.86	31.20	2.11

	Resource sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	16.78	7.01	6.73
CRI-rho	16.78	7.01	6.73
CRI-tau	16.78	7.01	6.73

	Time sensitivity		
	avg [%]	std dev [%]	skew [-]
CI	30.73	45.18	0.83
SI	32.33	37.68	1.10
SSI	5.08	9.23	3.09
CRI-r	11.22	9.66	3.04
CRI-rho	10.92	9.77	3.01
CRI-tau	11.22	8.35	1.30

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section “2.2. Risk Analysis”. Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy		
method - PF	MAPE [%]	MPE [%]
PV - 1	30.47	12.20
PV - SPI	21.71	13.10
PV - SCI	21.82	13.10
ED - 1	34.53	10.33
ED - SPI	21.71	11.33
ED - SCI	21.77	11.33
ES - 1	29.53	9.80
ES - SPI(t)	6.69	10.73
ES - SCI(t)	6.86	10.74

Simulated EAC accuracy		
method (PF)	MAPE [%]	MPE [%]
1	0.30	-28.65
CPI	0.54	-7.69
SPI	12.77	-7.60
SPI(t)	14.18	-33.09
SCI	12.81	-7.69
SCI(t)	14.21	-7.63
0.8 CPI + 0.2 SPI	3.99	-29.50
0.8 CPI + 0.2 SPI(t)	4.25	-0.91

According to the MAPE values¹ the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over 6 tracking periods. The Real Duration and Real Cost mentioned in section “2.3.3. Earned Value Management” are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

¹ The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	0	-48691.97	-206	1	0.72	0.50	0.85
std dev	0	44898.09	86.94	1	0.21	0.12	0.19
final	0	0.00	-152	1	1.00	0.73	1.00

2.3.3.2. Time forecasting

PD	52 Days
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Real Duration	71 Days
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Delay	36%
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EAC(t)			Real Accuracy	
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	64.33	11.94	15.96	9.39
PV - SPI	77.50	26.31	25.12	-9.15
PV - SCI	77.50	26.31	25.12	-9.15
ED - 1	73.33	9.75	11.27	-3.29
ED - SPI	87.67	20.69	24.41	-23.47
ED - SCI	87.67	20.69	24.41	-23.47
ES - 1	77.00	11.08	15.02	-8.45
ES - SPI(t)	107.33	21.46	51.64	-51.17
ES - SCI(t)	107.33	21.46	51.64	-51.17

2.3.3.3. Cost forecasting

BAC	197,472.31€
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Real Cost	197,472.31€
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On Budget	0%
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EAC			Real Accuracy	
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	197472.30	0.00	0.00	0.00
CPI	197472.30	0.00	0.00	0.00
SPI	248607.67	58818.63	25.89	-25.89
SPI(t)	290643.49	75186.33	47.18	-47.18
SCI	248607.67	58818.63	25.89	-25.89
SCI(t)	290643.49	75186.33	47.18	-47.18
0.8 CPI + 0.2 SPI	203487.08	5789.14	3.05	-3.05
0.8 CPI + 0.2 SPI(t)	206689.75	7481.44	4.67	-4.67